Master of Design
Year 1
Level 5

Semester 1
Sep–Dec 2026

Credit Value 3

Thursday 9.30– 12.30

Tutors:
Jen Yoohyun Lee
Henry Rui He
Mohana Das
Bharati Das

Lecture Rooms:
TU201
V322

Sub-cohort Rooms:
V1014 (TED)
V1122 (ISD)
V1123 (IBD)
V1124 (SSD)

3 Sept 2026

SD5008 Research and Analysis for Design

Synopsis

Design fundamentally aims to improve the human experience. Yet, uncritical interventions can frequently misfire, resulting in unintended ecological, social, or systemic harm. This course establishes that a truly effective design process must be rigorously mindful, contextually responsive, and deeply critical of its own impacts. Research, in this sense, is a foundational premise for such a design process. This course will equip you with the approaches needed to understand the dimensions of complex design contexts.

While research in general entails a systematic process of discovering the less-known and generating new, deeper insight, design research has a unique nuance that distinguishes it from both scientific research and humanities research. To grasp why we research as designers, it helps to see how design research stands apart from other academic traditions. Design research, by contrast to scientific and humanities research, is an applied, process-oriented endeavor intended to make the world a better place through human-centered action and material intervention. Rather than simply observing reality or testing a hypothesis, design research actively engages with the identified context to shape positive change. We will explore ways to engage with the world more effectively before conceptualizing and materializing design ideas.

By engaging with the concerned contexts, you will become aware of and develop skills to actively observe, listen, measure, explore, and collaborate with stakeholders to identify opportunities for meaningful intervention. Ultimately, you will learn how to grasp the critical insights needed to inform a rigorous, professional design process.

Keywords

Design research; Critical insight; Data collection; Analysis

Recommended Reading

Independent research and self-directed curiosity outside of class are key parts of your growth in this course, and exploring these texts will enrich your overall work and class discussions. The following texts are recommended.

- Bernard, H. R., & Ryan, G. W. (2010). *Analyzing qualitative data: Systematic approaches*. SAGE Publications.
- Blaikie, N. (2006). *Analyzing quantitative data: From description to explanation*. Sage Publications.
- Bryman, A., Bell, E., & Kleinknecht, S. W. (2022). *Social research methods*. Oxford University Press.
- Creswell, J. W. (2007). *Qualitative inquiry and research design: Choosing among five traditions* (2nd ed.). Sage Publications.
- Cross, N. (1999). Design research: A disciplined conversation. *Design Issues*, *15*(2), 5–10. https://doi.org/10.2307/1511837
- Denzin, N. K., & Lincoln, Y. S. (Eds.). (2018). *The SAGE handbook of qualitative research* (5th ed.). SAGE Publications.
- Frayling, C. (1993). Research in art and design. *Royal College of Art Research Papers*, *1*(1), 1–5.
- Friedman, K. (2008). Research into, by and for design. *Journal of Visual Art Practice*, *7*(2), 153–160. https://doi.org/10.1386/jvap.7.2.153_1
- Hair, J. F., & Babin, B. J. (2018). *Multivariate data analysis* (8th ed.). Cengage Learning.

- Koskinen, I., Zimmerman, J., Binder, T., Redström, J., & Wensveen, S. (2011). *Design research through practice: From the lab, field, and showroom*. Morgan Kaufmann.
- Olson, J. S., & Kellogg, W. A. (Eds.). (2014). *Ways of Knowing in HCI* (Vol. 2, p. 74). New York: Springer.
- Richards, L. (2021). *Handling qualitative data: A practical guide* (4th ed.). SAGE Publications.
- Tufte, E. R. (2018). *The visual display of quantitative information* (2nd ed.). Graphics Press.

Course Objectives and Learning Outcomes

The objectives of the course are:

- To develop students' appreciation for relationships between research, the practice of design, and scholarship on design.
- To prepare students for a practical and theoretical understanding of research planning and processes, including literature review, data gathering, analysis, and communication.
- To teach students to use and develop research knowledge through a pilot study project applicable to design at the graduate level.
- To help students develop awareness of research's value to design practice in the development of new artifacts, and its relationship to scholarship on design.

Upon completion of the subject, students will be able to:

- Discuss the role of research in design practice
- Justify the choice of appropriate research methods and their suitability to address a design issue or solve a design problem
- Justify the choice of appropriate research methods and data collection
- Utilize appropriate methods and tools to manage and analyze data
- Synthesize and present research findings using appropriate (re)presentation formats
- Understand, at a foundational level, design research as a professional specialization
- Communicate ideas effectively in written, oral, and visual formats
- Collaborate with peers, faculty, and stakeholders
- Continuously develop critical assessment of data in design practices
- Develop students' appreciation for relationships between research, the practice of design, and scholarship on design.

Course Components

The course includes three components:

- **Structured content lectures** in which the tutors will discuss topics related to design research and analysis for students' information and guidance in developing research.
- **Research study projects** in which groups of students will design and conduct a short study that includes a literature review, data collection, analysis, and communication. These projects must include a component of first-hand data gathering and/or generation, and the progress is to be demonstrated and communicated through presentations and the final report.
- **In-class individual written assessment** in which students will individually reflect on and articulate their in-depth understanding of research design at the end of the semester.

Because of the level of course enrolment, students are organized and expected to meet as follows:

- Students will meet for structured content lectures and research case lectures together as a **full cohort** (approximately 100 students).
- Students will meet for research study project presentation review in **sub-cohorts** of approximately 25 students.

- Tutors will divide students into **groups** of 3-4 students within each sub-cohort.
- Students will work with their group on their research study project.

Students will deliver their work on research study projects through the following deliverables (Please refer to the schedule below for delivery dates):

- **Research study project presentations**
  - Student groups present their progress on their research study project.
  - 5 ~ 15 minutes per group
  - Visual, in-person presentation (.pdf or .ppt) plus verbal presentation.
  - Must include written, visual, and verbal content developed to a professional standard.
  - Sections can be organized like the following, but not limited to:
    - Research Context and Background / Scope
    - Relevant Literature Review
    - Research Methods
    - Data and Analysis
    - Conclusions and next steps
  - Tutors will then give feedback on each research study project:
    - Overall progress and content
    - Research design and theorization
    - Thoroughness of background and literature review consideration
    - Appropriateness and execution of data gathering methods
    - Appropriateness and execution of analysis
    - Rigor and value of conclusions and insights drawn from the research

- **Research study report (Due 6 Dec 2026**)
  - A written document of approximately 5,000-8,000 words, in English
  - Submitted to plagiarism checking with report appended
  - Must include:
    - Images, tables, and diagrams as appropriate
    - Referencing citations in APA 7th Edition Referencing style
  - Each student group must make a digital copy submission of the following on Canvas:
    - PDF format. Under 10 Mb in size.
    - Filename: <year as two digits>_SD5008_<sub-cohort tutor surname>_<Research study project group number>_report.pdf
    - Example: 26_SD5008_Lee_07_report.pdf

Course Tutors

Jen Yoohyun Lee | SSD | jenn-y.lee@polyu.edu.hk

Dr. Jennifer Yoohyun Lee is a practitioner-researcher with a keen interest in exploring participatory processes, communication and agency, group decision-making, and social engagements through mixed methods approaches. She holds a Ph.D. in Design from PolyU. Previously educated at Goldsmiths, University of London (MA in Design) and Pratt Institute (BFA in Sculpture), Jen continues to practice and research in socially engaged art and design. Her research scope traverses urban ecology, participatory and community design, communication mechanisms, social media, systems thinking, (re)defining design problems, and more-than-human sensitivities.

Henry Rui He | ISD | khenryhe@gmail.com

Dr. Rui He is a Human-Computer Interaction (HCI) researcher and former Senior Game Engineer at NetEase Games. He earned his PhD in HCI from The Hong Kong Polytechnic University and holds an MSc with Distinction from The University of Hong Kong. His expertise spans HCI, generative AI, design creativity, and computer graphics, with publications in premier venues like

ACM UIST and CHI. A Gold-Level Unreal Authorized Instructor, Rui has taught game development and AI in design, and holds a PCT patent for game-engine film generation.

Mohana Das | TED | mohana.das@connect.polyu.hk
Dr. Mohana Das is a Visiting Lecturer in the School of Design at The Hong Kong Polytechnic University. She holds a PhD in Design from PolyU, a Master's in Architecture and Urban Design from Politecnico di Milano (110/110 cum laude) and a Bachelor's in Architecture from India. Her research employs mixed-methods approaches including surveys, in-depth interviews, focus groups, participatory action research and statistical analysis. A registered architect, her work examines how environments and technologies can support health, wellbeing and dignity for ageing populations and vulnerable communities. She has published in leading journals including the Journal of Urban Health and Frontiers in Public Health. She also serves as Senior Resident Tutor at PolyU's Student Halls of Residence.

Bharati Das | IBD | dr-bharati.das@connect.polyu.hk
Ms. Bharati Das is a multidisciplinary designer and researcher specializing in service design, branding strategies, design for entrepreneurship, and value creation. Her work lies at the intersection of AI-integrated pedagogy and cross-cultural management. Her doctoral research examines how psychographics can synthesize company and customer perceived values in service design. With an MDes in IBD from PolyU and an MBA from IIM Shillong, India, Ms. Das integrates design methodology with business strategy to build impactful innovation, backed by several Hong Kong entrepreneurship grants and over 4 years of industrial experience.

Language

Tutors will communicate, correspond, and teach in **English** as the primary medium of instruction according to University Policy. While informal use of other languages is permitted during casual conversations and peer collaboration in class, you are expected to actively participate in English during critiques as part of your assessment.

Use of GenAI

All uses of GenAI in support of your work and in-class discussions must be properly declared, including the specific GenAI used and the prompt used. Utilizing GenAI to supplement your learning is encouraged; however, any uses of GenAI that would complete your assignments without your critical input or hinder your learning in this class are not encouraged. If you are in doubt, please consult the tutors.

Class Participation

Students are required to attend all lectures and tutorials. To get the most out of this course, please keep the following expectations in mind:

- **Schedule & Presence:** We meet weekly on **Thursday mornings from 9:30 AM to 12:30 PM**. Fully engaging for the complete session ensures you stay connected with discussions and group work. Arriving more than 20 minutes late or leaving early without prior arrangement disrupts the class dynamic and may count as an unexcused absence.
- **Absences & Exceptions:** If you need to miss a class, please reach out in advance whenever possible. **Unplanned absences due to illness require a doctor's note to be officially excused. Accumulating three (3) unexcused absences may jeopardize** your ability to pass the course.
- **Impact on Assessment:** Attendance and active involvement contribute to your assessment. Meaningfully contributing to group tasks, posing thoughtful questions, and offering constructive feedback during peer presentations provides direct evidence of your critical thinking and engagement.

Add/Drop Dates and Procedures

The deadline for add/drop applications for Semester 1 2026-2027 is **12 September 2026.** Please be advised that after this date, you cannot apply to drop a course without penalty. To apply to add or drop a course, you must email the School of Design General Office before the deadline and state your intention to drop a course. Emails or other communication with tutors, discipline leadership, or other faculty will not suffice to add or drop a course and may affect your evaluation if your add/drop course paperwork is not received on time.

Assessment

Class Participation, Self & Peer Assessment (individual)	25%
In-class Individual Written Assessment (individual)	20%
Presentation (group)	25%
Research Report (group)	30%

Schedule

Week	Date & Place	Topic	Milestones
1	03/09/2026 TU201	**[Lecture]** Introduction to Design Research **[Special Lecture]** Prof. Jaden Park **[Tutorial]**	- Group arrangement after lecture
2	10/09/2026 V322	**[Library Workshop]** **[Lecture]** What & Why: Identifying and defining research dimensions; Research Through Design **[Tutorial]**	- Initial project idea / topic - Research purpose - Begin background & precedent review
3	17/09/2026 TU201	**[Lecture]** How #1: Approaches of Qualitative Research & Quantitative Research **[Tutorial]**	- Continue background review - Draft research question / creative proposition
4	24/09/2026 TU201	**[Lecture]** How #2: Data gathering mechanisms + AI **[Tutorial]**	-Submit **Research study project presentation 1** + Research topic + Literature mapping + Inquisitive direction + Research modalities in consideration
-	01/10/2026	*No Class*	- Initiate data gathering process
5	08/10/2026	**[Sub-cohort Meeting]** + Design of research modality [workflow] + Data / documentation	**Research study project presentation 2**
6	15/10/2026	**[Tutorial]**	- Adjustments to data collection mechanisms - Data collection
7	22/10/2026 V322	**[Lecture]** How #3: Sensemaking through data filtering and analysis – Descriptive & Analytical **[Tutorial]**	- Continue data collection - Begin data cleansing / editorial selection

8	29/10/2026	**[Sub-cohort Meeting]** + Updated design of research modality [workflow] + Data / documentation + Data filtering procedure	**Research study project presentation 3**
9	05/11/2026 V322	**[Lecture]** How #4 & So What: Communicating and developing design proposition **[Tutorial]**	- Begin data synthesis / analysis
10	12/11/2026	**[Sub-cohort Meeting]** + Updated data / documentation + Data filtering procedure + Data synthesis	**Research study project presentation 4**
11	19/11/2026	**[Tutorial]**	- Begin drawing findings
12	26/11/2026	**[Sub-cohort Meeting]** + Updated data / documentation + Data filtering procedure + Data synthesis + Newly gained insight	**Research study project presentation 5** - Begin drafting the final report
13	03/12/2026 TU201	**[In-class Individual Written Assessment]**	- Complete the research study report and submit by 6 December 2026

Grading Rubric

Grade & GPA	Class Participation & Peer Assessment — 25%	In-Class Individual Written Assessment — 20%	Group Presentation — 25%	Group Research Report — 30%
A+ (< 4.3) **A (<4.0)** **A- (<3.7)**	Exemplary class participation. Contributes substantially and proactively to course discourse and seminars. Peer evaluations evidence outstanding collaboration and positive leadership in group work.	Demonstrates exceptional understanding of design research principles. Written response is highly articulate, critically engaged, and shows insightful synthesis of course concepts.	Exceptional clarity, visual quality, and verbal delivery. Research aims, methodology, and outcomes are communicated seamlessly as a cohesive, well-coordinated team.	Rigorous, consistent, and valid research design. Demonstrates significant insight, strong methodological execution, and professional report formatting to a high standard.
B+ (<3.3) **B (<3.0)** **B- (<2.7)**	Good class participation. Contributes regularly and constructively to class discussions. Peer evaluations evidence positive and reliable teamwork with no significant issues.	Demonstrates strong understanding of core design research concepts. Written response is clear, coherent, and well-structured with good analytical depth.	Strong visual and verbal delivery. Effectively communicates research aims, processes, and outcomes with clear group coordination and engagement.	Method planning is clear, consistent, and well-aligned with research aims. Demonstrates solid analytical development, clear findings, and a well-organized report.
C+ (<2.3) **C (<2.0)** **C- (<1.7)**	Adequate class participation. Contributes occasionally or passively to seminars and presentations. Peer evaluations indicate minor negative feedback regarding group involvement.	Demonstrates basic or satisfactory understanding of course topics. Written response is functional but lacks critical depth, precision, or thorough reflection.	Satisfactory presentation standard. Communicates basic research elements, though visual design, verbal articulation, or group synthesis lacks polish.	Method planning is basic but satisfactorily aligned with aims. Research execution and report structure meet essential requirements but show limited analytical depth or insight.
D+ (<1.3) **D (<1.0)**	Poor class participation. Very limited contribution to course discourse. Peer evaluations highlight significant negative feedback regarding group participation.	Demonstrates poor or superficial understanding of research concepts. Written response is disorganized, vague, or falls below expected academic standards.	Poorly structured or rushed presentation. Weak verbal and visual delivery with unclear explanation of research processes or findings.	Methodological planning is unclear, inconsistent, or invalid for the stated aims. Report shows minimal research rigor, weak structure, and limited execution.
F (0.0)	Unexcused persistent absenteeism; complete failure to engage in class or contribute to group activities.	Failure to submit the assessment or work completely fails to meet basic academic requirements.	Failure to deliver the presentation within the designated schedule.	Failure to submit the research report or work fails to address course outcomes entirely.